

Overview of the BioDCASE 2026 Challenge on Bird Population Counting from Passive Acoustic Recordings

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Abstract—Passive acoustic monitoring is increasingly used for biodiversity assessment, yet estimating the number of individuals of a species, rather than merely detecting its presence, remains an open problem. The BioDCASE 2026 Bird Population Counting task addresses this in multi-species zoo aviaries, where the number of individuals of each species is known from curator records, providing a controlled but realistic benchmark for acoustic abundance estimation. Participants estimate the abundance of three target species (Greater flamingo, Red-billed quelea, Hadada ibis) from continuous 48 kHz recordings. The development set contains 140,899 three-second clips across six aviaries, and the evaluation set spans ten held-out aviaries (~380,000 clips), of which six are scored. An official baseline couples the ARIA detector with per-species detection-rate regression. Four teams submitted nine systems. The best reached a mean absolute error (MAE) of 9.17 individuals and the three top teams outperformed the baseline (MAE 24.50). The central finding is that detection-rate regression estimates the more separable quelea well but fails for the Greater flamingo, whose synchronous chorus saturates detection counts. The ibis is counted no better than a constant guess. The systems that improved most did so by modelling acoustic crowding directly, through synthetic-data density regression or a morphological measure of chorus energy, rather than by improving species detection.

Index Terms—bird population counting, passive acoustic monitoring, bioacoustics, abundance estimation, DCASE challenge

1. INTRODUCTION

Passive acoustic monitoring (PAM) has become a central tool for biodiversity assessment, offering a non-invasive and scalable way to survey vocalizing animals [1], [2]. Most acoustic pipelines answer a question of *presence*: given a recording, which species can be detected. A harder and less explored question is one of *abundance*, how many individuals are present, which is what drives most ecological decisions, from trend monitoring to intervention. Reliable acoustic abundance estimation would complement labour-intensive visual censuses and enable long-term monitoring from autonomous recorders.

Counting from sound is difficult because the mapping from vocal activity to individual count is neither one-to-one nor stationary. Individuals differ in call rate, some remain silent, and call rates vary with time and context [3]. Most importantly, when many conspecifics vocalize together the scene becomes a continuous chorus in which individual calls can no longer be resolved. In this regime raw detection counts *saturate*: beyond some density, additional individuals do not add proportionally more resolvable detections.

Zoo aviaries are the enabling choice for studying this problem. They preserve the acoustic phenomena that make wild counting hard, namely overlap, chorusing, and multi-species interference, while supplying the exact ground truth that wild sites cannot. This paper presents Task 6’s design, its multi-aviary dataset, and an official baseline. It then analyzes the four participating teams and draws out the central lesson, that flamingo chorus saturation is the dominant source of error, and that the systems which improved most did so by modelling acoustic crowding rather than by improving species detection.

2. RELATED WORK

We organize prior work by the route it takes from sound to abundance, noting for each what Task 6 adds.

Detectors and foundation models. Deep classifiers such as BirdNET [1] and the Perch bird-vocalization model [4], and ensembles built on them [5], make species recognition routine and are the natural front-end for counting. A confident detection is, however, a presence signal: how detections should be converted into an individual count is left unspecified, and this is precisely the gap Task 6 targets.

Cue-rate and vocal-activity models. Ecological cue-counting relates the number of vocal events to population size through a per-individual call rate [6], [7]. These models are interpretable and effective when calls are sparse and separable. They assume a stable, recoverable cue rate, an assumption with empirical support for sparse, separable colonies [8]. That assumption fails under synchronous chorusing, where the per-individual detectable rate falls as the colony grows. A similar decoupling of acoustic activity from abundance has been reported in dense seabird colonies [9]. The Greater flamingo instantiates this failure directly and makes it the pivot of the task.

Direct and density-based estimation. A complementary route estimates abundance from audio features without thresholded detection [10], echoing visual crowd counting, where density-map regression avoids individuating objects in dense scenes [11]. To our knowledge Task 6 is the first bioacoustic benchmark on which this route can be tested against detector-based counting under known counts, and the two leading systems adopt variants of it.

Benchmarking precedents. Prior public challenges targeted presence-type questions: binary bird-sound detection [12], species identification in soundscapes [13], and few-shot event detection [14]. Task 6 moves the target from “what is calling” to “how many are there,” pairing multi-species zoo soundscapes with exact, curator-recorded counts. Methods based on individual acoustic identity or spatial localization offer further routes, but require clean single-source audio or synchronized multi-channel deployments that a single-channel chorus precludes (Section 7).

3. TASK AND DATASET

3.1. Task definition

For each evaluation aviary and each present target species, a system outputs a single non-negative integer: the estimated number of individuals. The three species span complementary regimes (Table 1). The Greater flamingo (*Phoenicopterus roseus*) and Red-billed quelea (*Quelea quelea*) occur in large colonies that stress behaviour under saturation, while the Hadada ibis (*Bostrychia hagedash*) is a loud, distinctive caller present in very small numbers, so the same absolute error is proportionally far larger.

Systems are ranked by mean absolute error (MAE) over the evaluation aviaries, computed only over species present in each aviary. Root mean squared error (RMSE), coefficient of determination (R^2), and mean absolute percentage error (MAPE) are reported as supplementary metrics. MAE is the primary metric because it is in the natural unit of individuals, is less sensitive than RMSE to the extreme errors that saturation and extrapolation produce, and remains interpretable when averaged across species whose counts differ by

two orders of magnitude. It has a clear weakness: an error of three individuals is minor in a colony of 195 and severe in a group of four, and MAE does not express that difference. We retained MAE for comparability with the cue-counting literature and report MAPE alongside it. Section 5 shows this weakness is not hypothetical.

3.2. Data and ground truth

Audio was recorded in aviaries at seven European zoological facilities between 2025 and 2026. Facility identities are withheld to protect institutional privacy. Each aviary was recorded continuously for approximately one week, and a curated subset of representative days was released per deployment. Releasing every recorded day would have multiplied the dataset size without adding proportionate variety, and would have exposed the deployments from which some evaluation days are drawn. To avoid biasing the benchmark’s difficulty, the released days were selected automatically rather than by hand. We ran the baseline feature extraction (Section 4) over each full week-long recording and then searched candidate two-to-four-day subsets for the combination whose aggregate feature values most closely matched those of the whole deployment. The best-matching subset was released, so the data reproduces the full recording’s acoustic character, including its chorus and saturation structure. Recordings were segmented into consecutive, non-overlapping clips of approximately three seconds. All audio is stereo, 16-bit PCM WAV at 48 kHz, identical between the development and evaluation splits (Table 1).

Population counts were provided by the zoos as curator records. Because populations can change during a deployment, counts correspond to each aviary’s recording period. Every clip inherits its aviary’s count, so the target is the number of individuals housed there, not the number vocalizing in that clip. Three seconds is far too short for every housed individual to call, and vocal activity differs across species, so systems must aggregate evidence across the recording rather than treat any clip as carrying the full population.

The evaluation aviaries probe generalization along different axes for each species, and Table 1 gives the scored counts alongside the development ranges. The Greater flamingo, where saturation makes extrapolation the central question, is scored well outside its development range, at 11 birds below the development floor of 52 and at 195 above the ceiling of 161. The two scored quelea aviaries are further recording days from the same two deployments used in development, so their counts repeat the development values and test repeatability rather than transfer. The two scored ibis aviaries come from a facility not represented in development, so they do test transfer, but both hold four birds, at the development floor, leaving the count itself familiar. Section 5 reads each species accordingly.

Alongside the targets, each aviary houses many non-target species. For the evaluation set we released the union of all species across evaluation aviaries, a 67-species inventory, as a flat list rather than a per-aviary composition, since revealing composition would compromise the held-out design. Three caveats follow from this design and recur below. Development aviaries 5 and 6 are the same enclosure and population recorded twice, there are only two to four labelled aviaries per target species, and the species live in disjoint numeric regimes, so per-species modelling is effectively required.

4. BASELINE SYSTEM

The official baseline [15] is a two-stage, detector-plus-regression pipeline. In the first stage, the ARIA ensemble detector [5] labels every three-second clip with per-species detections and confidences. It combines BirdNET [1], the Perch classifier [4], and a fusion model over a zoo-specific 87-species inventory. A BirdNET-only front-end

Table 1: Task and dataset summary. Top: target species, development count ranges, and the counts of the scored evaluation aviaries. Bottom: split sizes; the scored evaluation set is the six aviaries containing the target species.

Target species	Counting regime	Dev. range	Eval. counts
Greater flamingo	Synchronous chorus; saturation	52–161	11, 195
Red-billed quelea	Dense flock; more separable	61–153	61, 153
Hadada ibis	Few, loud, distinctive callers	4–6	4, 4
Split	Aviaries / Hours $\approx$		Clips
Development	6 / 117		140,899
Evaluation (scored)	6 / 191		229,467
Evaluation (full)	10 / 317		$\sim$ 380,000

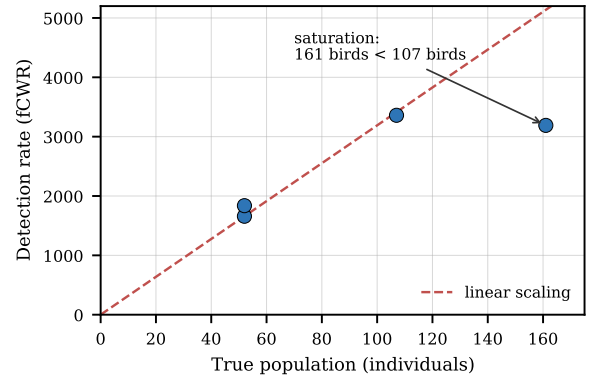


Fig. 1: Greater flamingo detection rate (fCWR) versus the number of individuals housed in each development aviary, which is not the number vocalizing at any moment. The 161-bird aviary falls below the 107-bird one, a direct signature of chorus saturation.

is also provided so the detector’s effect can be isolated. In the second stage, a feature builder aggregates detections to the aviary level, producing 80+ features spanning detection statistics, temporal-bout and occupancy structure, and optional scikit-maad acoustic and flock-calling indices [16]. A separate low-parameter estimator is fitted per species by leave-one-aviary-out (LOO) validation. Red-billed quelea uses a linear coefficient on the confidence-weighted detection rate (CWR), Hadada ibis a coefficient on vocal-bout rate, and Greater flamingo a flock-corrected CWR (fCWR) that rescales the detection rate by an acoustic-index factor intended to compensate for saturation. The estimator ships five interchangeable model families and reports the best per species, the choice depending on the upstream detector, with the two acoustic-index families skipped when audio is unavailable.

On the development set the ARIA baseline attains a LOO MAE of 11.5 individuals. On the evaluation set it attains 24.5, decomposing into flamingo 69.0, quelea 1.5, and ibis 3.0. The BirdNET-only front-end is weaker, reaching a development MAE of 18.4 and R^2 of 0.53 against ARIA’s 0.83. The error is concentrated almost entirely in the flamingo, where even fCWR cannot linearize the relationship. As Fig. 1 shows, the 161-bird development aviary yields a *lower* detection rate than the 107-bird one, a direct signature of saturation that upper-bounds any estimator built on detection counts.

5. SUBMITTED SYSTEMS AND RESULTS

Four teams submitted nine systems. All reused the baseline recipes for quelea and ibis and differed mainly in their treatment of the Greater

flamingo, confirming it as the task’s central difficulty. Table 2 reports all nine systems and the baseline.

Yang et al. (WHU) [17] reframe flamingo counting as density regression rather than detector post-processing. A convolutional-recurrent network (~ 1 M parameters) is trained to predict the number of flamingo calls per three-second fragment from a synthetic corpus of isolated Xeno-Canto flamingo calls mixed at *known counts* over real aviary backgrounds and non-target interferers, with source-level splitting to prevent leakage. Fragment predictions are then mapped to an aviary count by ordinary least squares. Supervising directly on synthetic counts sidesteps detector saturation. The system won on every metric (MAE 9.17; flamingo 20.0), gave the only flamingo estimate that does not collapse at either evaluation extreme, and ships code and trained weights. Its accuracy depends on how well the synthetic mixtures match real aviary acoustics, an assumption left unquantified, and it still overshoots the 11-bird aviary (40).

Leon et al. (UFAM) [18] introduce a detector-independent feature grounded in mathematical morphology [19]. After band-passing to the flamingo range (400–1500 Hz), the spectrogram is decomposed into structured and diffuse components, and the residual diffuse energy, the “stochastic band ratio,” quantifies the smeared chorus that arises under saturation. Multiplied by fCWR and fitted with a four-point linear model, it ranked second (MAE 13.33). The feature is interpretable and the entry is fully open-source. It still requires a same-species calibration set, and its largest error is an undershoot at 195 birds (155), where the feature itself begins to fold. It is the most accurate entry at 11 birds (18), and retuning the species whitelist raised its ibis error.

You (PKU) [20] submits four flamingo estimators that aggregate ARIA and BirdNET detection and acoustic statistics through low-parameter ridge regressions, hedging against saturation uncertainty. Quelea and ibis reuse the baseline recipes. A search over roughly 22,000 candidate flamingo models, ranked only on the four development points, produced near-zero development LOO error, but the selected models extrapolated poorly. The best variant scored 22.67 while others exceeded 100, and the 11-bird prediction (130) was the largest overshoot of any system.

Zhong (NJU) [21] estimates counts from stratified samples of each aviary under a strict validation gate that adopts a candidate only if it strictly reduces species-wise LOO error, otherwise reverting to the baseline. All attempted saturation, concurrency, and visibility corrections failed the gate, so the final system is a baseline fallback. Stratified sampling and partial detector coverage weakened the CWR features, leaving it below the full baseline (46.67). It is a transparent negative result rather than a new method.

The ranking tracks how much system escaped the detector for the flamingo. The quelea is handled well by the baseline recipe: systems that reused it on comparable detector inputs stayed at low single-digit MAE (baseline 1.5, You 3.5, Yang 4.5), whereas altering the detector or its inputs was costly, as Zhong’s sub-sampling shows (40.5).

The ibis column needs a different reading, and it is the clearest illustration of the metric limitation noted in Section 3. Both scored ibis aviaries hold four birds, and every system that reused the baseline recipe predicted one individual in each, giving MAE 3.0. A constant predictor fixed anywhere in the observed development range of four to six individuals would have scored better, between 0.0 and 2.0. The five entries at 3.0 are therefore not five systems agreeing; they are one recipe reproducing one fixed offset, which the bout-rate coefficient fitted on two development aviaries could not correct. Low absolute MAE on this species should not be read as accurate counting, and

Table 2: All nine submitted systems and the official baseline (reference row), sorted by evaluation MAE. Per-species MAE for Greater flamingo (Fla), Red-billed quelea (Que), and Hadada ibis (Ibi); Dev is the self-reported development LOO MAE. The Ibi column is discussed in the text: every entry at 3.0 predicted one individual where the true count was four.

Team	System	MAE	Fla	Que	Ibi	Dev
Yang	CRNN-Count	9.17	20.0	4.5	3.0	0.75
Leon	estoclin (S3)	13.33	23.5	6.0	10.5	2.25
Leon	estocpow (S2)	14.17	26.0	6.0	10.5	—
You	stack191	22.67	61.5	3.5	3.0	0.00
—	baseline	24.50	69.0	1.5	3.0	11.50
Leon	baseline (S1)	25.17	58.5	6.0	11.0	—
You	zoo252	31.33	87.5	3.5	3.0	—
You	aria56	40.33	114.5	3.5	3.0	0.00
Zhong	VG-ARIA	46.67	83.0	40.5	16.5	11.50
You	ext600	106.17	312.0	3.5	3.0	0.25

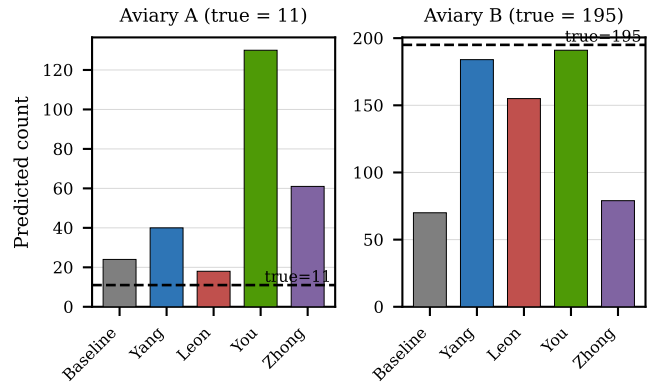


Fig. 2: Best-per-team Greater flamingo predictions at the two evaluation aviaries (true counts 11 and 195, dashed). Every system overestimates the low-density aviary, whereas several approach the high-density one.

the Hadada ibis is better described as unevaluated than as solved.

The ordering is thus decided almost entirely by the flamingo, whose per-species MAE ranges from 20 for the winner to 83 for the last ranked team. We analyze why in Section 6. Fig. 2 shows a second, distinct failure. At the high-density aviary several systems land within a few percent of the true 195, but every system, the baseline included, overestimates the 11-bird aviary, where a handful of confident detections already implies a substantial count and no method has nearby calibration points. Low-density extrapolation is thus a separate open problem from saturation, and the ibis result suggests it is not confined to the flamingo.

6. COMPARATIVE ANALYSIS

The submissions differ along a few axes that explain the results better than rank alone (Table 3).

Direct regression versus proxy-feature regression. Only Yang regresses counts directly, on synthetic supervision; the others regress a feature assumed monotone in abundance. Direct regression gave the most stable flamingo behaviour precisely because it never assumes proportionality between detections and callers, which is the assumption that saturation violates. A proxy feature can correlate strongly on the development aviaries, with Leon’s combined feature reaching Pearson $r = 0.998$ over four aviaries, yet break where its monotonicity fails, as at the 195-bird evaluation aviary. The distinction matters beyond this challenge: it separates methods whose error is bounded by the detector’s front-end from methods that can, in principle, exceed it.

Table 3: Core assumptions of the four systems. “Proxy” denotes regression on a feature assumed monotone in abundance rather than a direct count target.

Axis	Yang	Leon	You	Zhong
Detector dependence	low (bypass)	partial	high	high
Count target	direct	proxy feat.	proxy feat.	baseline
Representation	learned CRNN	morphology	detector stats	detector stats
External data	yes	no	no	no
Saturation handling	density regr.	diffuse-energy feat.	hedged search	none (gated)

Bypassing the detector for the count does not mean dispensing with it. A deployed system must still establish which species are present, so the practical target is a detector-informed estimator that identifies the species by detection but does not read abundance off detection rates.

Handcrafted and learned representations converge. Leon’s morphological ratio and Yang’s learned network are two routes to the same physical insight, that a dense chorus produces a diffuse acoustic *texture*, distinct from resolvable calls, that grows with the number of simultaneous vocalizers. That two independent approaches, one interpretable and detector-free, one high-capacity and data-driven, both outperform detection-counting is the strongest evidence in the challenge that the informative signal lies in the texture of overlap rather than the count of events. The handcrafted route is transparent and cheap to compute, requiring no training and only a band-pass filter and a morphological decomposition per clip, but it needs same-species calibration. The learned route generalizes better across the density range but depends on the realism of its synthetic training distribution.

External data, capacity, and practicality. External audio was used only by Yang, and only to synthesize count-labelled training data under source-level leakage control. It was decisive, making direct count supervision possible without real per-fragment labels, but carries an unquantified domain-match assumption. Capacity helped only when coupled to a representation that addresses saturation. You’s search, lacking such a representation, minimized four-point error without transferring, whereas the same budget spent on a density-aware representation produced the best system. The productive axis is the observation model of overlap, not raw capacity.

Small-sample fragility. The full scoreboard makes the statistical fragility of the task concrete. You’s four variants span evaluation MAE 22.7–106.2 despite near-identical, near-zero development error, and Leon’s three cluster at 13.3–25.2. Within-team variance from model choice therefore rivals between-team differences, so a rank is partly a matter of which calibration was submitted. The leaderboard’s top is consequently firmer than its middle: the winner alone avoids collapse at either density extreme.

7. DISCUSSION, LIMITATIONS, AND FUTURE WORK

The challenge separates acoustic bird counting into two problems. Counting detections handles sparse, separable callers, and the quelea reached low error with a one-parameter recipe. Modelling acoustic crowding remains the open frontier and dominated every score. Crucially, this divide is set by vocal behaviour, not by colony size. The Red-billed quelea occupies a numeric range comparable to the flamingo, 61–153 against 52–161 individuals, yet its detection counts scale with population while the flamingo’s saturate. The most plausible reason is that quelea calls remain temporally separable while the flamingo’s synchronous vocalizations merge into a continuous chorus. Saturation is thus a property of call synchrony and overlap rather than of abundance, which is why no single detection-rate model serves both species.

Several limitations bound what the benchmark shows. Only two to four labelled aviaries exist per species, so every count model is calibrated on very few points, and development LOO error proved a fragile model-selection signal, with You’s search the clearest example. The zoo setting preserves overlap and chorusing but is not fully wild, so leaderboard performance should not be read as wild-transfer performance. Because some evaluation aviaries continue development enclosures, the two splits are not fully independent, and only the flamingo is scored outside its development range (Section 3). Results are dominated by that one species, whose two out-of-range aviaries drive most of the ranking, while the quelea errors measure repeatability and the ibis is scored at a single familiar count. The lessons of Section 6 rest on four independent submissions and are indicative rather than conclusive. Finally, vocal activity need not equal abundance for intermittently vocal individuals.

The most promising directions are grounded in these outcomes. First, uncertainty-aware, calibrated counting. Given how weakly development error predicted evaluation error, future editions should score predictive intervals and calibration rather than point MAE alone. Second, a metric that reflects proportional error, and a constant-count predictor scored per species as a reference row. The ibis result shows that a species can post the best absolute score in the table while being counted worse than a constant, and a relative or logarithmic criterion, reported per species rather than pooled, would make effort spent on small populations visible. Third, establishing whether the two effective ideas are complementary or redundant. Section 6 argues that the morphological feature and the learned density regressor read the same underlying quantity, the textural signature of overlap, so a naive combination may add parameters without information. At the number of calibration colonies available here, that distinction decides whether fusion improves or over-fits, and should be tested rather than assumed. Fourth, ecologically constrained synthetic data with source-level leakage control, and domain adaptation from aviaries to wild soundscapes, since naive augmentation would not reproduce Yang’s gains. Longer term, individual-level acoustic identity [22] and multi-channel spatial arrays offer routes beyond aggregate proxies, and semi-supervised pipelines could relax the labelled-aviary bottleneck.

8. CONCLUSION

BioDCASE 2026 Task 6 established acoustic population counting as a tractable but unsolved problem, and contributes the first public benchmark pairing multi-species zoo soundscapes with exact, curator-recorded counts. The best system reached a cross-species MAE of about nine individuals, beating the baseline by more than 60%, and simple per-species detection-rate regression already handles a dense but separable caller. The open frontiers are chorus saturation, which upper-bounds any method built on detection counts, and extrapolation to densities outside the calibration range, which every system got wrong at the low end. Both trace to the same root, the small number of labelled colonies that leaves calibration fragile.

The challenge’s central lesson generalizes beyond this dataset. Progress came not from better species detection but from changing what is regressed: the leading systems succeed by modelling the acoustic *texture* of a crowd, whether learned from synthetic mixtures or measured by spectrogram morphology, rather than by counting resolvable events. Acoustic abundance estimation thus becomes the design of an observation model for overlapping vocal activity, a framing that connects the task to cue-rate theory in ecology and that should transfer to other chorusing taxa. The dataset and baseline are publicly available [15], [23] to support this work.

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